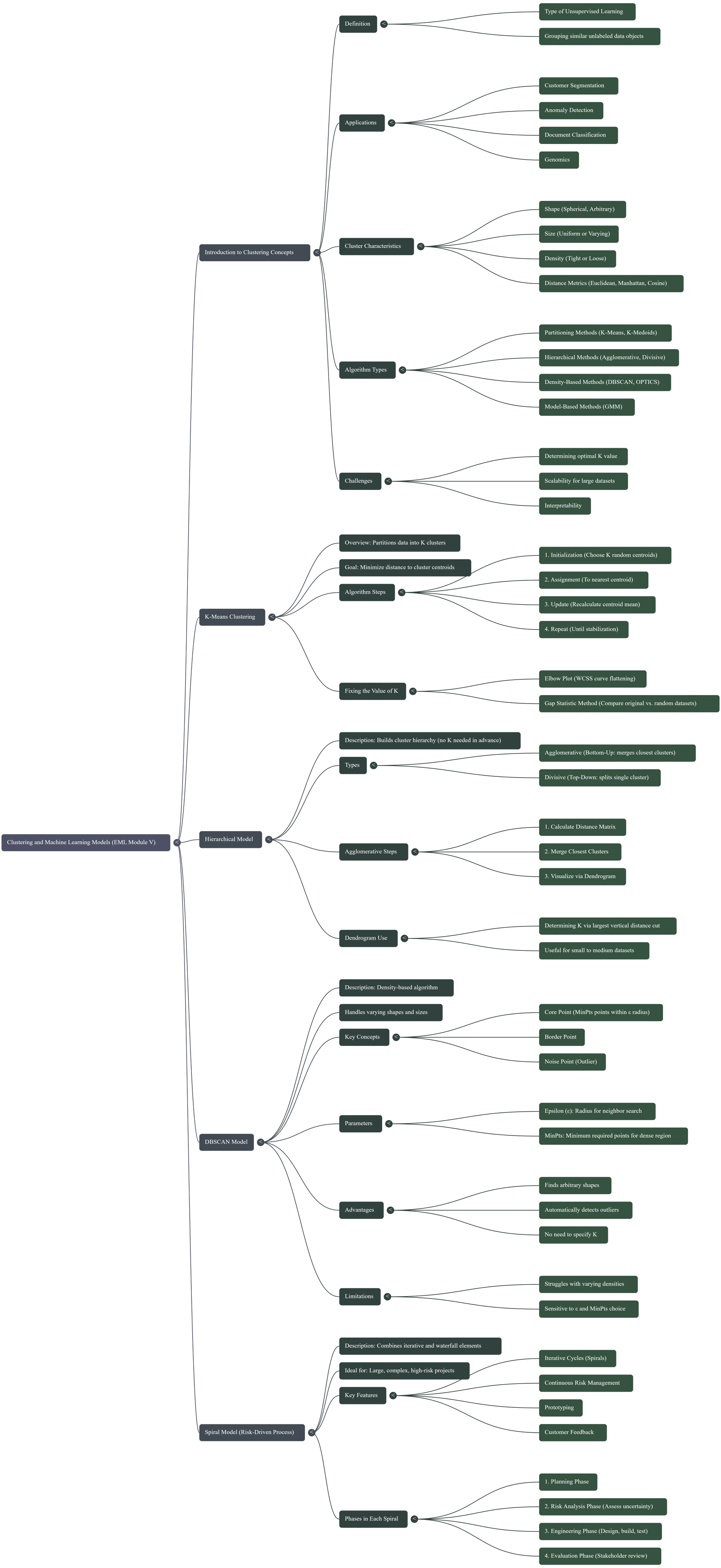

Clustering and Machine Learning Models (EML Module V)
Introduction to Clustering Concepts
Definition
Type of Unsupervised Learning
Grouping similar unlabeled data objects
Applications
Customer Segmentation
Anomaly Detection
Document Classification
Genomics
Cluster Characteristics
Shape (Spherical, Arbitrary)
Size (Uniform or Varying)
Density (Tight or Loose)
Distance Metrics (Euclidean, Manhattan, Cosine)
Algorithm Types
Partitioning Methods (K-Means, K-Medoids)
Hierarchical Methods (Agglomerative, Divisive)
Density-Based Methods (DBSCAN, OPTICS)
Model-Based Methods (GMM)
Challenges
Determining optimal K value
Scalability for large datasets
Interpretability
K-Means Clustering
Overview: Partitions data into K clusters
Goal: Minimize distance to cluster centroids
Algorithm Steps
1. Initialization (Choose K random centroids)
2. Assignment (To nearest centroid)
3. Update (Recalculate centroid mean)
4. Repeat (Until stabilization)
Fixing the Value of K
Elbow Plot (WCSS curve flattening)
Gap Statistic Method (Compare original vs. random datasets)
Hierarchical Model
Description: Builds cluster hierarchy (no K needed in advance)
Types
Agglomerative (Bottom-Up: merges closest clusters)
Divisive (Top-Down: splits single cluster)
Agglomerative Steps
1. Calculate Distance Matrix
2. Merge Closest Clusters
3. Visualize via Dendrogram
Dendrogram Use
Determining K via largest vertical distance cut
Useful for small to medium datasets
DBSCAN Model
Description: Density-based algorithm
Handles varying shapes and sizes
Key Concepts
Core Point (MinPts points within ε radius)
Border Point
Noise Point (Outlier)
Parameters
Epsilon (ε): Radius for neighbor search
MinPts: Minimum required points for dense region
Advantages
Finds arbitrary shapes
Automatically detects outliers
No need to specify K
Limitations
Struggles with varying densities
Sensitive to ε and MinPts choice
Spiral Model (Risk-Driven Process)
Description: Combines iterative and waterfall elements
Ideal for: Large, complex, high-risk projects
Key Features
Iterative Cycles (Spirals)
Continuous Risk Management
Prototyping
Customer Feedback
Phases in Each Spiral
1. Planning Phase
2. Risk Analysis Phase (Assess uncertainty)
3. Engineering Phase (Design, build, test)
4. Evaluation Phase (Stakeholder review)